「Def.

Let G be a group, and $A \subseteq G$ be a non-empty subset.

Define $C_G(A) \overset{\text{def}}{=} \{g \in G \mid gag^{-1} = a \text{ for all } a \in A\}$, which is called the Centralizer,

and $N_G(A) \overset{\text{def}}{=} \{g \in G \mid gAg^{-1} = A\}$, which is called the Normalizer.

「Prop. Let G be a group, and $A \subseteq G$ be a non-empty subset, and $H \leq G$.

1). $N_G(A)$ and $C_G(A)$ are subgroup of G.

2) The normalizer $N_G(H)$ is the largest subgroup such that H is normal.

3). A subgroup $H \leq G$ satisfies $H \leq C_G(H)$ if and only if H is abelian